

Element Notes UI Specification

Overview

Element Notes allow collectors to add personal context to each item in their collection: acquisition date, source, price paid, condition, storage location, and free-form notes. Notes are private by default and accessible via a quick drawer/modal from any element card.

Core Features

1. Quick Access

- **Trigger:** Single click/tap on “Add Notes” button on element card
- **Location:** Accessible from:
 - Cabinet view (periodic table element cards)
 - Collection grid view (owned items)
 - Wishlist view (for planning/tracking)

2. Seven Note Fields

Field	Type	Purpose	Required
Acquisition Date	Date	When you acquired this element	No
Source	Text	Where you bought it (shop, gift, trade)	No
Price Paid	Currency	How much you paid	No
Condition	Select	Condition rating (Mint, Good, Fair)	No
Storage Location	Text	Where you store it (cabinet shelf, safe)	No
Notes	Textarea	Free-form personal notes	No
Private	Toggle	Visibility control (private by default)	Yes

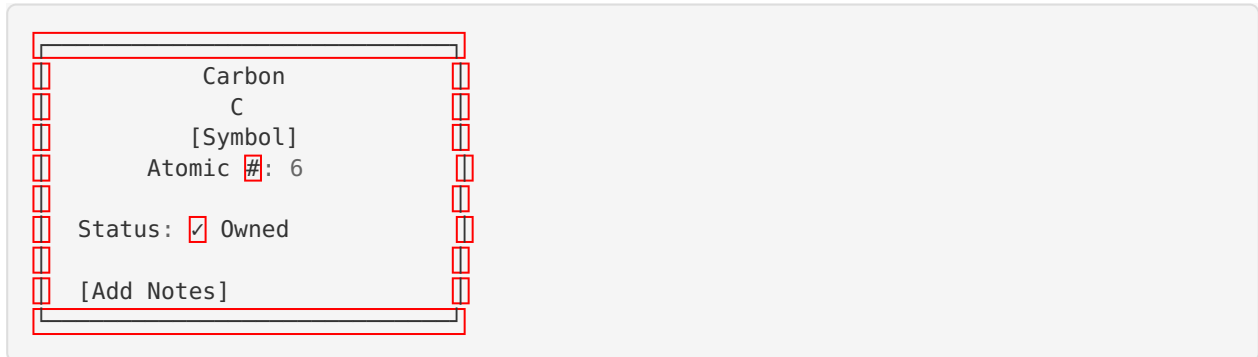
3. Privacy

- **Default:** All notes are private (only visible to owner)
- **Future:** Optional sharing with other collectors (Phase 3+)

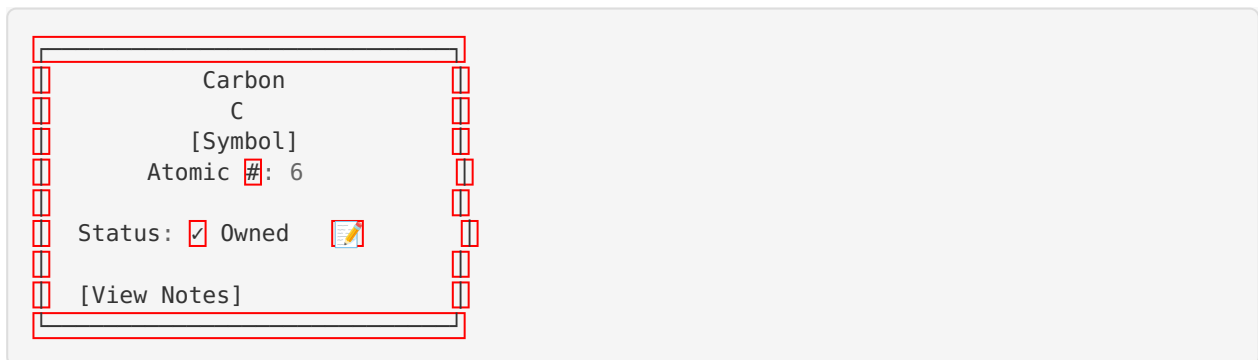
UI Wireframes

A. Element Card with Notes Indicator

Without Notes (Default State)



With Notes (Indicator Shown)



Indicator:  icon in top-right or next to status badge when notes exist

B. Notes Drawer (Desktop)

Trigger: Click "Add Notes" or "View Notes" on element card

Layout: Slide-in drawer from right side (400px wide)

Cabinet View

H

He

Li

Be

B

C

[More elements...]

Carbon (C) Notes [x Close]

Personal Details

Acquisition Date

May 15, 2024 [17]

Source

Luciteria.com

Price Paid

\$ 24.99

Condition

Mint ▼

Storage Location

Display Cabinet, Shelf 2

Notes

First element **in** my collection!
Gift from Sarah **for** birthday.

Privacy

☒ Keep these notes private

[Cancel] [Save Notes]

Last updated: 2 days ago

C. Notes Drawer (Mobile)

Trigger: Tap “Add Notes” on element card

Layout: Full-screen modal (slides up from bottom)

The image shows a mobile app interface for 'Carbon (C) Notes'. The form includes the following fields and elements:

- Back** button and **Carbon (C) Notes** title.
- Personal Details** section with a pencil icon.
- Acquisition Date** field with a date picker showing **May 15, 2024** and a calendar icon.
- Source** field with the text **Luciteria.com**.
- Price Paid** field with the text **\$ 24.99**.
- Condition** field with a dropdown menu showing **Mint**.
- A scrollable area with the text **[Scroll for more fields...]**.
- [Cancel]** and **[Save Notes]** buttons at the bottom.

Mobile Optimizations:

- Full-screen overlay (not drawer)
- Bottom action bar with Cancel/Save buttons
- Larger touch targets (48px minimum)
- Native date picker on tap
- Auto-save on field blur (optional)

D. Empty State (No Notes Yet)

The image shows the empty state of the 'Carbon (C) Notes' app. The interface includes the following elements:

- Carbon (C) Notes** title and **[x] Close** button.
- A pencil icon representing a note.
- The text **No notes yet for this element**.
- Instructions: **Add details about when and where you acquired this piece, or any personal notes you'd like to keep.**
- A button labeled **[Add Notes]**.

After clicking “Add Notes”, the form appears with all fields empty.

Field Specifications

1. Acquisition Date

Type: Date picker

Format: MM/DD/YYYY (or user’s locale)

Behavior:

- Click field → opens calendar dropdown
- Can type date manually
- Default: empty (not pre-filled)
- Validation: Cannot be future date
- Helper text: “When did you acquire this element?”

UI:

Acquisition Date

May 15, 2024

[ 17]

When did you acquire this element?

2. Source

Type: Text input (single line)

Max Length: 100 characters

Behavior:

- Autocomplete suggestions based on previous entries:
 - * “Luciteria.com”
 - * “Gift from [name]”
 - * “eBay”
 - * “Local shop”
 - * “Trade”
- Case-insensitive
- Trimmed on save

UI:

Source

Luciteria.com

Where did you get it?

[Autocomplete dropdown appears **if** matches found:]

✓ Luciteria.com (used 15 times)
Luciteria Store (used 3 times)

3. Price Paid

Type: Currency input

Format: \$ XX.XX (USD default, future: multi-currency)

Behavior:

- Numeric keyboard on mobile
- Auto-formats to 2 decimal places
- Optional (collectors may not want to track)
- Accepts integers or decimals
- Strips non-numeric characters

Validation:

- Min: \$0.01
- Max: \$999,999.99 (arbitrary upper bound)
- Error if negative or invalid

UI:

Price Paid

\$ 24.99

How much did you pay? (optional)

States:

- Empty: Shows placeholder "\$ 0.00"
- Typing "24.99" → auto-formats to "\$ 24.99"
- Invalid input (e.g., "abc") → shows error "Please enter a valid price"

4. Condition

Type: Select dropdown

Options:

- **Mint** - Perfect condition, no flaws
- **Good** - Minor wear, fully functional
- **Fair** - Visible wear, some imperfections

- **Poor** - Significant damage or degradation
- **Unknown** - Condition not assessed

Default: Empty (no pre-selection)

UI:

Condition

Mint ▼

[Dropdown expanded:]

- ☒ Mint
- ☐ Good
- ☐ Fair
- ☐ Poor
- ☐ Unknown

Visual Indicators (shown on card when condition set):

- Mint: ★ (gold star)
- Good: ✓ (green check)
- Fair: ⚠ (yellow warning)
- Poor: ● (red circle)
- Unknown: ? (gray question mark)

5. Storage Location

Type: Text input (single line)

Max Length: 100 characters

Behavior:

- Autocomplete based on previous entries
- Common suggestions:
 - * "Display Cabinet, Shelf X"
 - * "Safe"
 - * "Desk drawer"
 - * "Storage box"
- Helps collectors remember physical location

UI:

Storage Location

Display Cabinet, Shelf 2

Where do you store this element?

6. Notes

Type: Textarea (multi-line)

Max Length: 1,000 characters

Behavior:

- Free-form text entry
- Line breaks preserved
- Markdown NOT supported in v1 (plain text only)
- Character counter shows remaining space

UI:

Notes

First element in my collection!
Gift from Sarah for my birthday.
Sentimental value.

987 / 1000 characters remaining

Use Cases:

- Personal memories (“First element I ever bought”)
- Historical context (“Purchased during 2024 sale”)
- Future plans (“Want to upgrade to 50mm version”)
- Research notes (“Used in chemistry demo”)

7. Privacy Toggle

Type: Checkbox

Default: ☒ Checked (private)

Behavior:

- When checked: Notes visible only to owner
- When unchecked: Notes visible to other collectors (Phase 3+ feature)
- In Phase 2: Always private (toggle shown but grayed out)
- Tooltip: “Public sharing coming soon!”

UI:

Privacy

☒ Keep these notes private

[Phase 3+ when unchecked:]

☐ Keep these notes private

 Your notes will be visible to other collectors viewing this element. Personal data like price will remain private.

Interaction Flows

Flow 1: Adding Notes to New Element

User owns Carbon (C)

- Clicks "Add Notes" on Carbon card in cabinet
- Notes drawer slides in from right (desktop) or up (mobile)
- Form shows empty fields
- User fills out:
 - Acquisition Date: May 15, 2024
 - Source: Luciteria.com
 - Price Paid: \$24.99
 - Condition: Mint
 - Storage Location: Display Cabinet, Shelf 2
 - Notes: "First element in my collection!"
 - Privacy: ☒ Checked
- User clicks "Save Notes"
- Drawer closes, success toast appears: "Notes saved for Carbon"
- Carbon card now shows  indicator
- **Next** time user clicks "View Notes", form pre-filled with saved data

Flow 2: Editing Existing Notes

```
User has notes for Copper (Cu)
├─ Clicks "View Notes" on Copper card
├─ Notes drawer opens with pre-filled data
├─ User edits "Price Paid" from $15.00 to $18.00
├─ User clicks "Save Notes"
├─ Drawer closes, success toast: "Notes updated for Copper"
└─ "Last updated" timestamp refreshed
```

Flow 3: Deleting Notes

```
User wants to remove notes for Iron (Fe)
├─ Opens "View Notes" for Iron
├─ Clicks "Delete All Notes" button (bottom of form, red text)
├─ Confirmation dialog appears:
│   └─ "Are you sure? This cannot be undone."
│       └─ [Cancel] [Delete Notes]
├─ User clicks "Delete Notes"
├─ Notes record deleted from database
├─ Drawer closes, toast: "Notes deleted for Iron"
└─ Iron card no longer shows  indicator
```

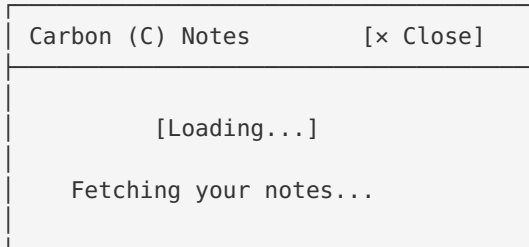
Flow 4: Quick Save (Auto-save on blur - Optional Feature)

```
User edits notes for Silver (Ag)
├─ Opens notes drawer
├─ Changes "Condition" from "Good" to "Mint"
├─ Clicks into "Notes" field (blur event on Condition)
├─ System auto-saves Condition change in background
├─ Small checkmark appears briefly: ✓ Saved
└─ User can close drawer without clicking "Save" (already saved)
```

Note: Auto-save is optional enhancement. For MVP, use explicit "Save Notes" button.

Component States

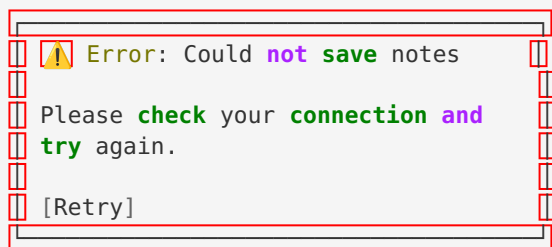
1. Loading State



2. Saving State

[Save Notes] button changes to:
[Saving...] (grayed **out**, spinner icon)

3. Error State



4. Success Toast



Technical Specifications

Database Schema

ElementNote Model

```

model ElementNote {
  id                String    @id @default(cuid())
  userId            String
  collectionItemId  String    @unique

  acquisitionDate   DateTime?
  source            String?
  pricePaid         Decimal? @db.Decimal(10, 2)
  condition         Condition?
  storageLocation   String?
  notes             String?   @db.Text
  isPrivate         Boolean   @default(true)

  createdAt         DateTime @default(now())
  updatedAt         DateTime @updatedAt

  user              User      @relation(fields: [userId], references: [id])
  collectionItem    CollectionItem @relation(fields: [collectionItemId],
references: [id])

  @@index([userId])
}

enum Condition {
  MINT
  GOOD
  FAIR
  POOR
  UNKNOWN
}

```

Key Points:

- One note record per collection item (1:1 relationship)
- `collectionItemId` is unique constraint (ensures one note per item)
- `pricePaid` stored as Decimal for accuracy
- All fields except `isPrivate` are optional
- `updatedAt` auto-updates on save

API Endpoints

GET `/api/notes/:collectionItemId`

Purpose: Fetch notes for a specific collection item

Response:

```
{
  "id": "note_abc123",
  "userId": "user_xyz789",
  "collectionItemId": "item_def456",
  "acquisitionDate": "2024-05-15T00:00:00Z",
  "source": "Luciteria.com",
  "pricePaid": 24.99,
  "condition": "MINT",
  "storageLocation": "Display Cabinet, Shelf 2",
  "notes": "First element in my collection! Gift from Sarah.",
  "isPrivate": true,
  "updatedAt": "2024-05-27T14:32:00Z"
}
```

404 Response (no notes exist):

```
{
  "error": "No notes found for this item"
}
```

POST /api/notes

Purpose: Create new notes for collection item

Request Body:

```
{
  "collectionItemId": "item_def456",
  "acquisitionDate": "2024-05-15",
  "source": "Luciteria.com",
  "pricePaid": 24.99,
  "condition": "MINT",
  "storageLocation": "Display Cabinet, Shelf 2",
  "notes": "First element in my collection!",
  "isPrivate": true
}
```

Response: Same as GET response

PUT /api/notes/:noteId

Purpose: Update existing notes

Request Body: Same as POST (partial updates allowed)

Response: Updated note object

DELETE /api/notes/:noteId

Purpose: Delete notes

Response:

```
{  
  "success": true,  
  "message": "Notes deleted"  
}
```

Validation Rules

Field	Validation
acquisitionDate	Must be valid date, not in future
source	Max 100 chars, trimmed
pricePaid	Numeric, 0.01 - 999,999.99
condition	Must be one of enum values
storageLocation	Max 100 chars, trimmed
notes	Max 1,000 chars
isPrivate	Boolean, defaults to true

Performance Considerations

Lazy Loading: Notes only loaded when drawer opened (not pre-fetched for all elements)

Caching: Cache notes locally in browser for 5 minutes to avoid repeated API calls

Debouncing: If implementing auto-save, debounce saves by 1 second to avoid excessive API calls

Optimistic Updates: Update UI immediately on save, then sync with backend. Rollback if save fails.

Accessibility (a11y)

Keyboard Navigation

- **Tab:** Navigate between fields
- **Enter:** Submit form (when focus on Save button)
- **Escape:** Close drawer without saving
- **Shift+Tab:** Navigate backward

Screen Reader Support

- Label all form fields with `<label>` tags
- Use `aria-label` for icon buttons
- Announce success/error toasts with `role="alert"`

- Drawer has `role="dialog"` and `aria-modal="true"`

ARIA Labels

```
<button aria-label="Add notes for Carbon">Add Notes</button>
<input aria-label="Acquisition date" type="date" />
<textarea aria-label="Personal notes about this element"></textarea>
```

Mobile-Specific Considerations

Touch Targets

- Minimum 48×48px for all interactive elements
- Larger padding around form fields
- Bottom action bar for easy thumb reach

Input Types

- `type="date"` → Native date picker
- `type="number"` → Numeric keyboard for price
- `inputmode="decimal"` → Decimal keyboard

Scrolling

- Drawer content scrolls independently
- Fixed header with element name
- Fixed footer with action buttons
- Content area scrolls in between

Future Enhancements (Post-MVP)

Phase 3+ Features

1. **Photo Uploads:** Attach photos of the element to notes
2. **Public Sharing:** Share notes with community (controlled by privacy toggle)
3. **Note Templates:** Pre-filled templates for common scenarios
4. **Export:** Download all notes as PDF/CSV
5. **Markdown Support:** Rich text formatting in notes field
6. **Voice Notes:** Record audio notes
7. **Tags:** Custom tags for categorization (e.g., “gift”, “rare”, “upgrade target”)

Design System

Colors

- **Primary:** #667eea (action buttons)
- **Success:** #48bb78 (save confirmation)

- **Error:** #f56565 (validation errors)
- **Gray:** #718096 (helper text)

Typography

- **Form Labels:** 14px, medium weight, gray-700
- **Input Text:** 16px, regular weight, black
- **Helper Text:** 12px, regular weight, gray-500
- **Character Counter:** 12px, regular weight, gray-400

Spacing

- Form field vertical spacing: 24px
- Label to input gap: 8px
- Button spacing: 12px
- Drawer padding: 24px

Summary

Purpose: Personal context for collection items

Access: Single click from element card

Fields: 7 (all optional except privacy toggle)

Privacy: Private by default

UI: Drawer (desktop) / Full-screen modal (mobile)

Philosophy: Quick, unobtrusive way to add personal meaning to collections without disrupting browsing flow.